Mohamed Medhat Kamal

Senior Software Engineer · Node.js · NestJS · TypeScript · Laravel | AI & LLM Integrations

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SUMMARY

Senior Software Engineer with 12+ years specializing in backend architecture, high-throughput APIs, and scalable distributed systems using Node.js, NestJS, TypeScript, and PHP/Laravel. Deep focus on core backend fundamentals: clean database modeling, query optimization, Redis caching strategies, asynchronous queues, and idempotent operations. Proven track record improving system performance on platforms serving 1M+ active users, cutting API response times by 60%, and maintaining 90%+ test coverage. Experienced in real-time telemetry (WebSockets, MQTT) and building practical production AI LLM agents using Anthropic SDK, LangGraph.js, RAG, and vector databases.

SKILLS

Backend & Architecture: Node.js, NestJS, Next.js (App Router / Server Actions), Express.js, TypeScript, Prisma ORM, RESTful APIs (OpenAPI/Swagger), WebSockets, MQTT, PHP 8+ / Laravel, Microservices, Multi-Tenant Architecture (Database-per-Tenant Isolation)

AI & LLM Engineering: LangGraph.js, LangChain.js, Claude API (Anthropic SDK), RAG (chunking, embeddings, PostgreSQL pgvector + HNSW), structured outputs (Zod), tool calling & deterministic code guardrails, automated LLM evals, Ollama

Frontend: React 19, Next.js 16, TypeScript, Vue 3 (Composition API, Pinia), Vuetify, Tailwind CSS, shadcn/ui, GSAP

Databases & Messaging: PostgreSQL (pgvector, pg-boss queues, automated pg_dump backups), MySQL, Redis (caching, pub/sub, queues), MongoDB

Testing & Quality: Vitest (600+ unit & DB-backed integration tests), Jest, Supertest, PHPUnit, TDD, CI regression gates (90%+ coverage)

DevOps, Cloud & Observability: Docker, Caddy, Nginx, CI/CD (GitHub Actions), Linux, AWS (EC2, RDS, S3, ELB), GCP, Sentry, DataDog, Bugsnag

Security & Domain Expertise: OAuth 2.0, Auth.js / NextAuth v5 (JWT), RBAC, CASL | Fintech & double-entry digital wallets, 1M+ user mobility platforms, Multi-Tenant ERP (ZATCA, Decimal.js), Industrial IoT (MES/OEE/Andon), Document Automation

EXPERIENCE

Senior Software Engineer · Business Development Services · ERP & SmartX IIoT · Remote — Jun 2026 – Oct 2026

- Re-architected BDS's core enterprise ERP (**Next.js 16, React 19, TypeScript, Prisma 7, PostgreSQL 16**) into a multi-tenant platform (**prisma-platform**) with strict database-per-tenant isolation and automated daily **pg_dump** backup pipelines supporting branches across Egypt and Saudi Arabia (ZATCA QR invoicing).
- Engineered a high-throughput background calculation engine (**pg-boss** worker queues, **Decimal.js**, and streaming **ExcelJS / TUS** chunked uploads) that pre-computes financial snapshots across 450k+ ledger rows, cutting dashboard load times to 0.1–0.4s.
- Stabilized critical financial, inventory, and import workflows by authoring **633 automated tests in Vitest** (234 unit tests + 399 database-backed integration tests) and enforcing strict role-based data redaction.
- Architected and finalized the **SmartX Industrial IoT** manufacturing platform (MES, CMMS, OEE tracking, and Andon boards) and ported its real-time alert escalation engine to **NestJS + TypeScript**.
- Built the factory-to-cloud telemetry pipeline using **MQTT** and **WebSockets**, connecting custom C++ ESP32 industrial gateways and AI vision cameras on the shop floor to multi-tenant cloud dashboards.

Stack: Next.js 16, React 19, NestJS, Node.js, TypeScript, Prisma 7, PostgreSQL 16, pg-boss, Vitest (633 tests), TUS, Tailwind CSS, MQTT, WebSockets, Docker, C++ (ESP32)

Senior Full-Stack Engineer · Zimam (Imtithal Platform, 1M+ users · via Squadio) · Remote — Dec 2025 – Jun 2026

- Architected a two-way RMS/OMS platform synchronization engine keeping employee records, RBAC roles, and credentials consistent across distributed services, paired with an idempotent CLI reconciliation command for zero-drift recovery.
- Built a unified cloud storage abstraction layer across local disk, Google Cloud Storage, and AWS S3 using signed URLs, a secure file-streaming controller, and backward-compatible legacy path resolution.
- Engineered an automated DOCX-to-PDF legal document pipeline (unoserver) generating four regulated contract types (Contract, NDA, Custody, Handover) with dynamic template variables, digital signature cropping, Hijri calendar dates, and content-hash caching.
- Designed and shipped the end-to-end Application Invitations workflow (schema migrations, REST APIs, transactional emails, and UI) plus the public TypeScript + Vue 3 + GSAP candidate portal for Hajj/Umrah workforce onboarding.
- Delivered custody management, shift scheduling, training quizzes, and dynamic ID badge generation modules; partnered directly with the PM and GM on sprint planning and release architecture.

Stack: TypeScript, Node.js, Vue 3, Vuetify, GSAP, PHP 8.3 / Laravel, MySQL, Redis, Docker, GCP, AWS S3

Senior Backend Engineer · Telgani (Mobility Marketplace, KSA, 1M+ users · via Squadio) · Remote **Feb 2024 – Nov 2025**

- Designed and scaled high-concurrency RESTful APIs serving mobile applications with 1M+ active users across Saudi airports and cities.
- Co-architected the core payments and digital wallet engine (wallets, loyalty points, vouchers, dynamic discount rules) using strict double-entry bookkeeping, atomic rollback mechanisms, and zero-discrepancy financial audit trails.
- Cut average API response latency by 60% through SQL query refactoring, composite indexing, and multi-layer Redis caching strategies.
- Drove API-first engineering with full OpenAPI/Swagger specifications, SOLID domain refactoring (improving maintainability by 40%), and 90%+ automated test coverage across unit and integration suites.
- Collaborated directly with the CTO, Polish engineering lead, and mobile/QA squads on system design, zero-downtime releases, and production observability (DataDog, Bugsnag).

Stack: TypeScript, Node.js services, PHP / Laravel, MySQL, Redis, AWS (EC2, RDS, S3, ELB), Docker, WebSockets, DataDog, Bugsnag

Full-Stack Engineer / Team Lead · Squadio (Enterprise Client Engineering, KSA) · Remote **Dec 2022 – Jan 2024**

- **Aklat:** Promoted from Full-Stack Engineer to Team Lead, owning technical architecture and code reviews; resolved critical security and performance bottlenecks, integrated Moyasar payments (STC Pay, Apple Pay, MADA) and Odoo ERP, and increased subscription conversion by 35%.
- **Eat-Hub:** Architected the complete backend API from system design to cloud deployment, built the SPA admin dashboard, and shipped real-time WebSocket chat between customers and restaurants.
- **Soiree Portal:** Designed dual API backends (mobile + restaurant web portal) with real-time WebSocket notifications and live analytics dashboards.

Stack: Node.js, TypeScript, Vue.js, Laravel, MySQL, Redis, WebSockets, Docker, AWS, Odoo ERP API

Full-Stack Engineer & Team Lead · Mobile World (Conversational AI & Chatbots, UAE) · Remote **Apr 2021 – Nov 2022**

- Led a team of 5 engineers and introduced Agile Scrum workflows, accelerating production release cycles from 1 month to 1 week.
- Architected a low-code Bot Builder Platform and delivered 15+ high-throughput WhatsApp Business and chatbot engines across the MENA region with sub-200ms webhook response latency.
- Built a reusable frontend component library that reduced new client dashboard build times by 50%.

Full-Stack Software Engineer · Z Novation & Early Career (Venue Manager NYC) **Jun 2013 – Mar 2021**

- Engineered 10+ production web and desktop applications, including **Venue Manager (NYC, USA)** (hospitality booking, bottle-service automation, and POS APIs), **Tenant & SalyJolie ERPs** (cross-platform **Node.js + Electron.js** desktop accounting and warehouse software with 6+ financial ledgers), and custom CLI code generators that reduced CRUD boilerplate by 70%.

EDUCATION

B.Sc. Information Technology (Web Development) · Advanced Academy, Giza, Egypt

LANGUAGES

Arabic (native) · English (professional)

LIVE PROJECTS

Open source: AI Real-Estate Assistant (Node.js · TypeScript · LangGraph.js · PostgreSQL pgvector · 22-case eval gate · 643 Vitest API tests) · github.com/m-medhat-dev/real-estate-agent

www.telgani.com · phommunity.com

Video demos: youtube.com/playlist?list=PL-7vd1W-gHEb6ThtC9e9-Uxb6gO-f9xef